

AI Crypto Advisor – Assignment Overview

Project Overview

AI Crypto Advisor is a fullstack web application that serves as a personalized crypto investor dashboard. The application allows users to sign up, log in, complete a short onboarding quiz, and receive a daily dashboard tailored to their selected crypto assets, investor type, and content preferences.

The dashboard always includes the four required sections: Market News, Coin Prices, AI Insight of the Day, and Fun Crypto Meme. User preferences are used to personalize the content inside these sections rather than hiding or removing sections.

Architecture & Implementation

The project is built with a React + Vite frontend and a Node.js + Express backend. The backend uses SQLite with Prisma as the ORM, and authentication is implemented with JWT and bcrypt password hashing.

The backend exposes REST API endpoints for authentication, onboarding, dashboard data, and voting feedback. User preferences are stored in the database after onboarding and are later used to generate personalized dashboard content.

The frontend communicates with the backend through API helper functions and stores the JWT token in localStorage. Protected pages such as onboarding and dashboard are accessible only after login.

Personalization Logic

Personalization is based on three onboarding inputs: selected crypto assets, investor type, and preferred content types.

Selected assets affect coin prices, market news, memes, and AI insights. Investor type affects the wording and focus of the AI insight. Content preferences influence the focus of the AI insight and the type of fallback or ranked market news shown to the user.

Coin prices are fetched using the CoinGecko API. Market news can use CryptoPanic if an API key is available, with static fallback data when no key is provided. AI insights can use OpenRouter when configured, and fallback insights are provided when no AI API key is available. Memes are served from static data for reliability.

Feedback and Future Model Improvement

Each dashboard section includes thumbs up/down voting. Votes are stored in the database with the user ID, section, item ID, and vote value. A unique constraint prevents duplicate votes for the same user and item, allowing future votes to update the existing feedback record.

In a future production version, this feedback could be used to improve personalization by learning which content types, assets, topics, and formats each user prefers. Over time, the system could rank news, insights, and memes based on user feedback patterns and potentially train or fine-tune a recommendation model using aggregated feedback data.

AI Tools Usage

During the assignment, I used AI tools such as ChatGPT and Claude to support planning, architecture review, debugging, and deployment guidance. I used these tools as assistants for reasoning and code review, while manually reviewing, testing, and validating the final implementation decisions.